Predicting Autonomy Levels in AI-for-Science Projects

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Summary

- **Question.** Can a short project description predict the autonomy level (L1–L5) assigned to an AI-for-Science project?
- **Completed.** Data preparation (320 projects), comparison of five classifiers, tuning, nested cross-validation, one held-out test, and error analysis. New this week: the discipline experiment you suggested.
- **Main result.** The SVMs performed best on the held-out test (Macro-F1 0.65–0.66; majority baseline 0.11). Most errors involve L3 (human-in-the-loop) and the L4/L5 boundary.
- **Discipline experiment.** Adding scientific discipline to the description made all five models slightly worse. The discipline field turned out to be very sparse.
- **Next.** Freeze the model and evaluate it once on newly added inventory projects as an independent test.

1 Data

The data come from an inventory of publicly documented AI-for-Science projects that I help build in separate research work. Each project has a short description and a level: L1 traditional ML, L2 foundation models, L3 human-in-the-loop AI, L4 agentic AI, L5 autonomous AI scientist or self-driving lab.

From 338 named projects, I removed 12 with multiple levels, 1 without a level, and 5 without a description, leaving **320 projects**. I deleted explicit level mentions (e.g., "L3", "level 3") from the text. I had planned to merge levels into two classes if some were too small, but every level had enough examples, so I kept all five.

Table 1: Projects per level, with a fixed 80/20 stratified split.

	L1	L2	L3	L4	L5	Total
Development	97	37	34	47	41	256
Test	24	9	8	12	11	64

L1 is 38% of the data, so a model could look accurate by mostly predicting L1. The labels come from the inventory and were not independently verified for this study, so the experiment measures agreement with existing labels, not the true autonomy of each system.

2 Method

- **Features.** TF-IDF on words and two-word phrases. TF-IDF and the classifier are in one scikit-learn `Pipeline`, so the vocabulary is learned only from the training part of each fold. This prevents information leaking from validation or test data.
- **Models.** Dummy baseline (always L1), logistic regression, KNN, decision tree, and, following your suggestion to compare decision functions, both a linear SVM and an RBF SVM.
- **Evaluation.** All development work used five-fold stratified cross-validation on the 256 development projects. The 64 test projects were used only once, at the end. I chose the primary model *before* looking at the test set.
- **Measures.** Macro-F1 is the main measure because it weights each level equally, so the small L3 class counts as much as L1. I also report accuracy and Cohen's kappa (linearly weighted kappa where noted, which treats an L1/L5 confusion as worse than an L4/L5 confusion).

3 Results of the Text Experiment

Model selection. After grid search, RBF SVM had the best development score (Macro-F1 0.492). Nested cross-validation (inner search, outer evaluation) gave a less optimistic estimate and RBF SVM was again highest (0.460). I fixed RBF SVM as the primary model at this point. The top models were close, so I do not claim RBF is clearly better.

Decision-tree complexity. Increasing the minimum leaf size from 1 to 20 made the tree simpler, but validation Macro-F1 dropped from 0.489 to 0.341. The training–validation gap shrank only because training performance fell. A simpler tree did not generalize better here.

Table 2: Held-out test results (64 projects). Kappa is linearly weighted.

Model	Macro-F1	Accuracy	Kappa
RBF SVM (primary)	0.645	**0.719**	**0.761**
Linear SVM	**0.662**	0.703	0.743
Logistic regression	0.603	0.672	0.726
KNN	0.468	0.531	0.477
Decision tree	0.451	0.516	0.538
Dummy (always L1)	0.109	0.375	0.000

- All models beat the baseline. The Dummy reaches 37.5% accuracy while recognizing only L1, which is why Macro-F1 was the main measure.
- The two SVMs were best. RBF got 46 of 64 correct, Linear SVM 45. Linear SVM had slightly higher Macro-F1, but I kept RBF as decided before testing rather than switching after seeing the result.
- Test scores were higher than development scores. This is not a real improvement: the final models used more training data, and with 64 projects the test scores are uncertain.

Where the model fails. Correct predictions per level: L1 23/24, L2 7/9, **L3 2/8**, L4 6/12, L5 8/11. L3 was hardest, and L4 and L5 were often confused with each other. I read all of these error cases. Typical patterns:

- **Automation is not autonomy.** *Culture Biosciences* (L3, predicted L5) is described as "remote

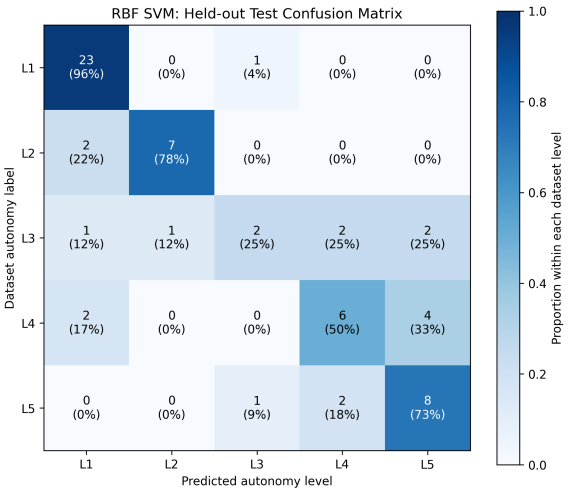


Figure 1: RBF SVM on the test set. Rows are inventory labels, columns are predictions.

automated fermentation / cell-culture experiments." The text says the experiments are automated, but not who decides what to run next, which is exactly what separates L3, L4 and L5.

- **Same word, different meaning.** In *Fear-Neuro-Inspired Reinforcement Learning* (L3, predicted L5), "autonomous" refers to self-driving cars, not autonomous science. A bag-of-words model cannot tell the difference.
- **Not every error is explained by the text.** *Autonomous quantum-sensing experiments* (L5, predicted L4) clearly describes autonomous decisions, yet the model still predicted L4.

These are likely explanations, not proven causes. I did not change any label based on model predictions.

4 New Experiment: Does Discipline Help?

Following your feedback, I tested whether a project's scientific discipline adds information. To change only one factor, I kept the same 256 projects, the same five folds, the same TF-IDF, and each model's tuned settings, and compared three inputs: **text only**, **discipline only** (one-hot encoded, missing values as `Unknown`), and **text + discipline**. As a check, the text-only scores exactly reproduced my earlier tuning results.

Table 3: Mean five-fold Macro-F1. The last column groups rare disciplines into one category.

Model	Text only	Discipline only	Text + discipline	Text + grouped discipline
Logistic regression	0.485	0.285	0.432	0.464
Linear SVM	0.463	0.280	0.409	0.465
RBF SVM	0.492	0.278	0.429	0.483
KNN	0.445	0.190	0.420	0.323
Decision tree	0.489	0.285	0.449	0.450

- **Discipline alone has weak signal**: about 0.28 Macro-F1, above the baseline (0.11) but far below text.
- **Adding it to text hurt every model**: Macro-F1 fell by 0.025–0.063 and kappa by 0.021–0.066. The drop appeared in three or four of the five folds for every model.

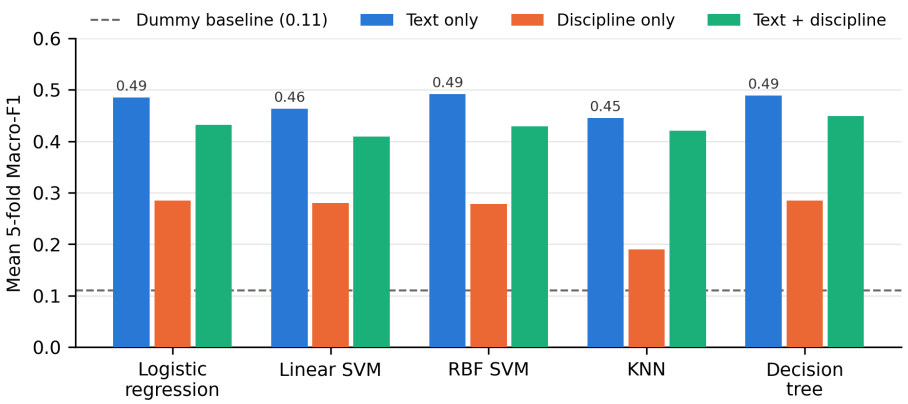


Figure 2: Mean five-fold Macro-F1 on the development projects.

- **Likely reason**: the 256 projects have 139 different discipline values, and 110 appear only once (e.g., "materials science; chemistry" is its own category). Also, descriptions often already name the field.
- **Follow-up check**: grouping disciplines seen fewer than five times removed most of the drop for logistic regression and the SVMs, but still gave no gain over text alone. KNN got much worse, showing that distance-based models are sensitive to how feature blocks are combined.

Conclusion: with the discipline field as currently recorded, it adds no useful information beyond the description. These are exploratory results on development data and are not comparable to the test-set scores above.

5 Next Steps

1. **Freeze the final model**: text only, RBF SVM. No further tuning on the old test set.
2. **Independent test**: collect projects added to the inventory after my snapshot, remove overlaps with the original 320 (renamed systems, rewritten descriptions, related projects), and evaluate the frozen model once.
3. **Final report**: compare the new errors with the L3 and L4/L5 patterns found here.

6 What I Learned

- **Baselines and the right measure matter.** A model that only predicts L1 gets 37.5% accuracy; Macro-F1 exposes it.
- **Ideas must be tested, not assumed.** A simpler tree and an extra feature both seemed reasonable, and neither helped.
- **Evaluation boundaries matter.** Fitting preprocessing inside each fold, choosing the model before testing, and planning a separate independent test keep the results honest.
- **Errors teach more than scores.** Reading misclassified cases showed that short descriptions often omit who makes decisions, the key information for this taxonomy.

Limitations. Small data (only 8 L3 test projects), unequal classes, short descriptions, labels not independently verified, and an inventory that is not a random sample of all AI-for-Science work.